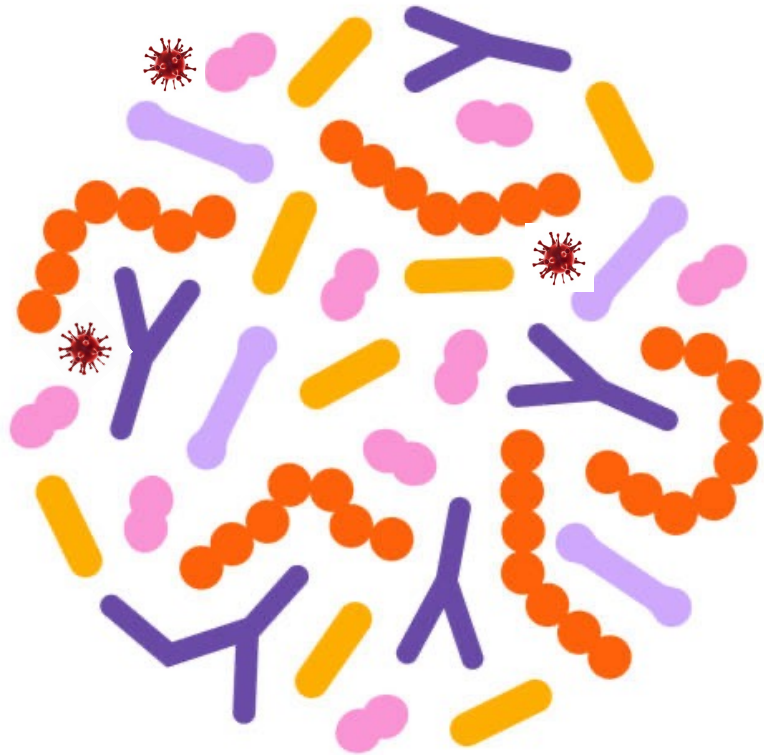




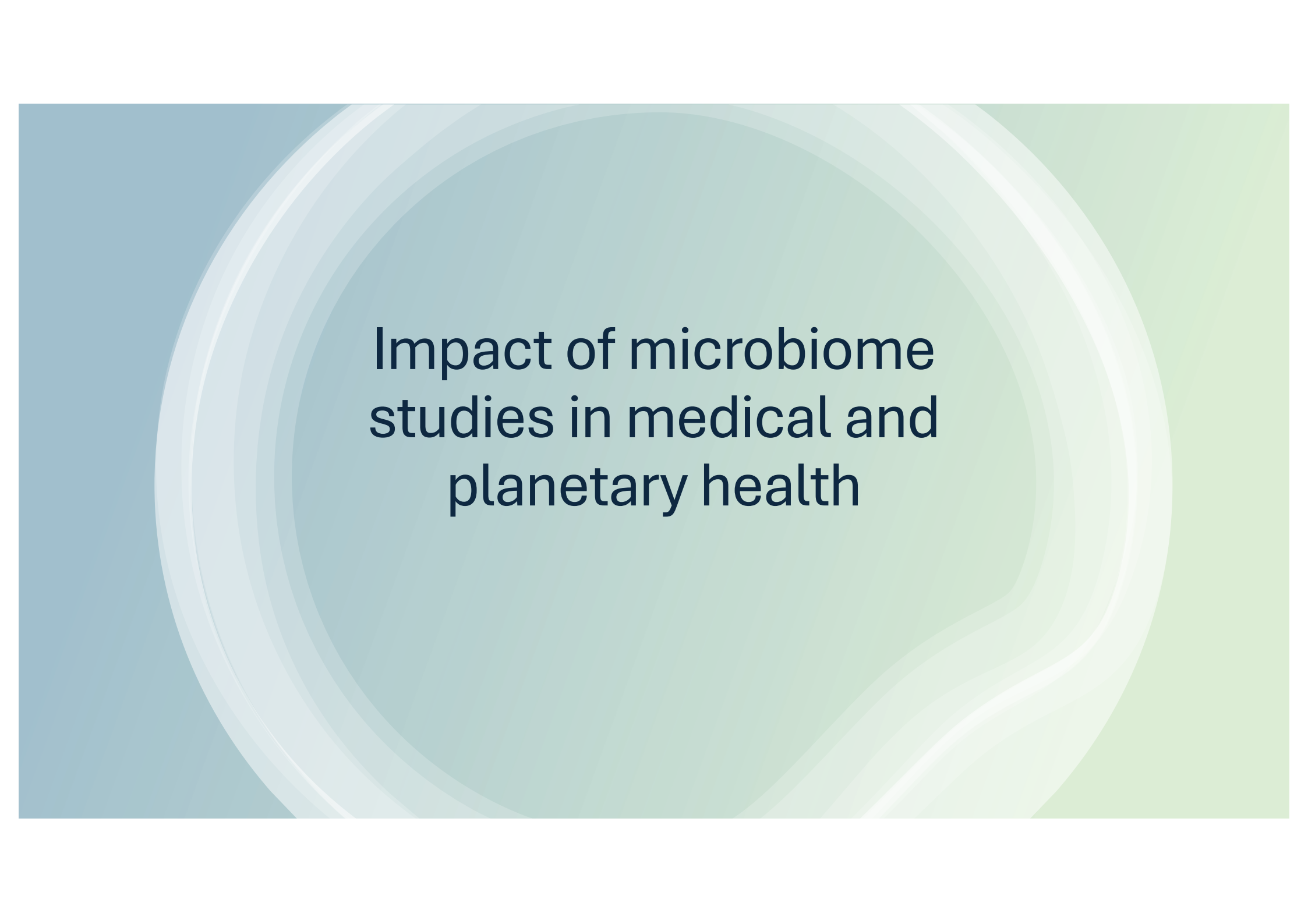
Introduction to Metagenomics

Data Science platform

BRIGHT

Spring Data-type Courses

6th May 2026



Impact of microbiome studies in medical and planetary health

Role of the intestinal microbiome

Disease modifier

Cardiometabolic disorders

- Obesity
- Insulin resistance
- Type 2 diabetes mellitus
- NAFLD/NASH
- Atherosclerosis
- Hypertension
- Idiopathic thrombocytopenic purpura

Gastrointestinal disorders

- Clostridium difficile infection
- Ulcerative colitis
- Crohn's disease
- Chronic pouchitis
- Irritable bowel disease
- Idiopathic constipation
- Antibiotic resistant pathogens

Neuropsychiatric disorders

- Encephalopathy
- Autism spectrum disorders
- Multiple sclerosis
- Parkinson's disease
- Chronic fatigue syndrome
- Neurodegenerative disorders
- Appetite disorders
- Depression & anxiety

Immunologic disorders

- Graft-versus-host disease
- Multiple organ dysfunction syndrome
- Colonic cancer
- Arthritis
- Allergy
- Eczema
- Celiac disease

Microbiome has been associated with a variety of diseases

- Microbiota: cause or consequence of the disease?
- For some diseases it may also be a disease modifier or may metabolize the drugs administered to the diseased individuals

Causal relationship

Obesity associated gut microbiome – a transmissible trait

An obesity-associated gut microbiome with increased capacity for energy harvest

Peter J. Turnbaugh, Ruth E. Ley, Michael A. Mahowald, Vincent Magrini, Elaine R. Mardis & Jeffrey I. Gordon 

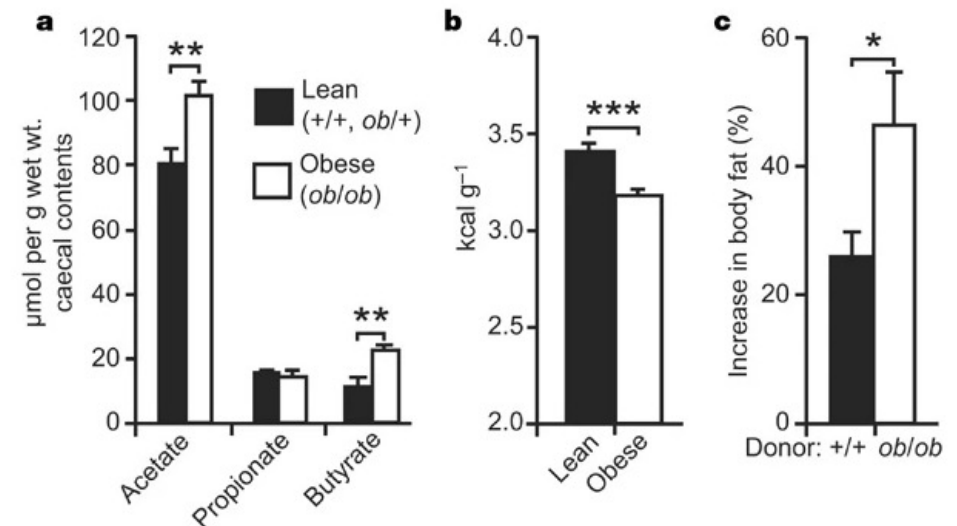
Nature 444, 1027–1031 (2006) | [Cite this article](#)

135k Accesses | 9059 Citations | 1196 Altmetric | [Metrics](#)

<https://doi.org/10.1038/nature05414>

- Obese mice had an increased capacity to harvest energy from the diet
- **The trait is transmissible!**
Colonization of germ-free mice with an ‘obese microbiota’ resulted in a significantly greater increase in total body fat than colonization with a ‘lean microbiota’

Figure 3: Biochemical analysis and microbiota transplantation experiments confirm that the *ob/ob* microbiome has an increased capacity for dietary energy harvest.



Genetically obese mice had an increased concentration of the major fermentation end-products: butyrate and acetate

They also had significantly less energy remaining in their faeces

Germ-free mice colonized with an obese mice microbiota exhibited significantly greater increase in body fat

The Human Microbiome Project (HMP)

Article | [Open access](#) | Published: 13 June 2012

Structure, function and diversity of the healthy human microbiome

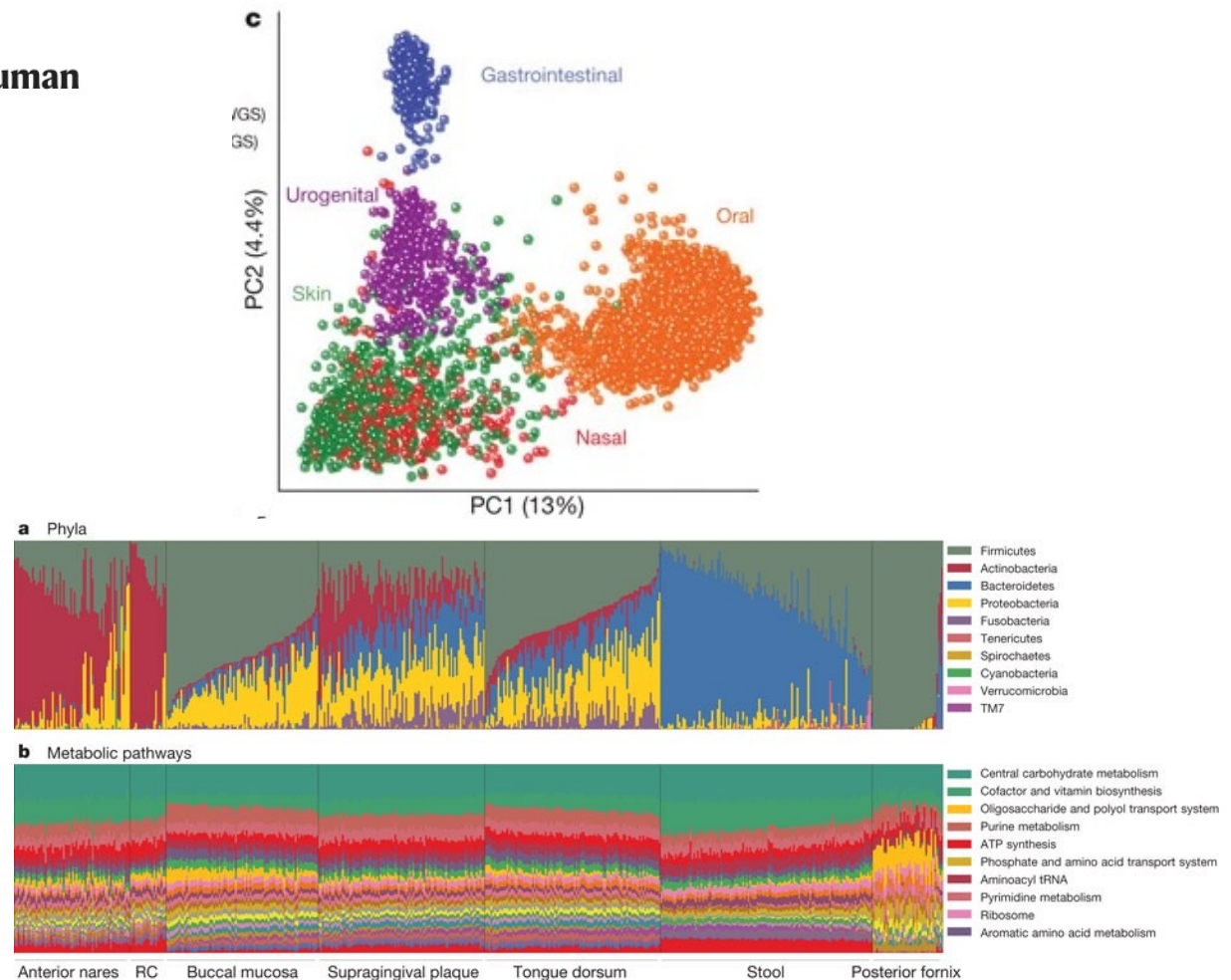
[The Human Microbiome Project Consortium](#)

[Nature](#) **486**, 207–214 (2012) | [Cite this article](#)

450k Accesses | 1240 Altmetric | [Metrics](#)

<https://doi.org/10.1038/nature11234>

- Great initial resource to study the microbiome (based on thousands of samples)
- Diversity of the human microbiome is concordant among measures, unique to each individual, and strongly determined by microbial habitat
- Carriage of microbial taxa varies while metabolic pathways remain stable within a healthy population



Richness of human gut microbiome correlates with metabolic markers (lessons of MetaHIT study)

Article | Published: 28 August 2013

Richness of human gut microbiome correlates with metabolic markers

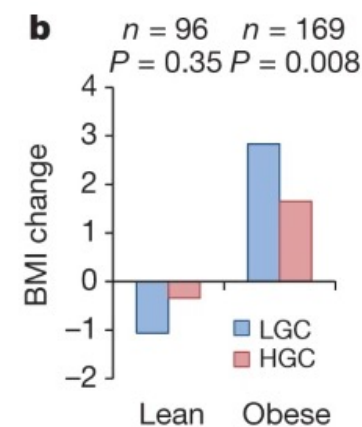
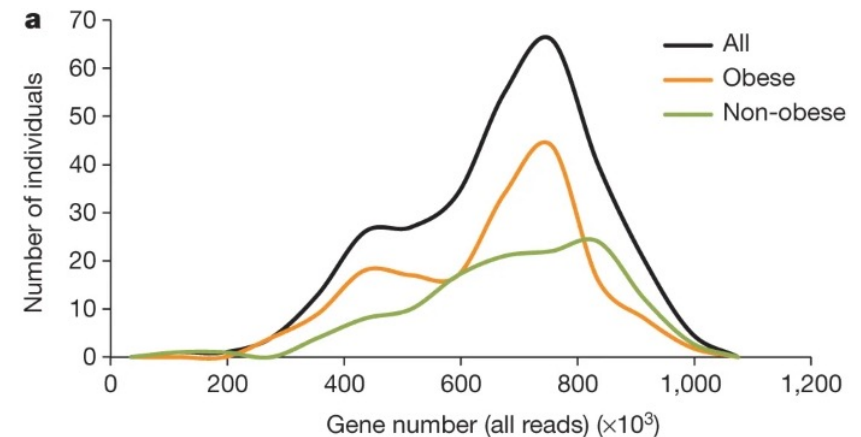
[Emmanuelle Le Chatelier](#), [Trine Nielsen](#), [Junjie Qin](#), [Edi Prifti](#), [Falk Hildebrand](#), [Gwen Falony](#), [Mathieu Almeida](#), [Manimozhiyan Arumugam](#), [Jean-Michel Batto](#), [Sean Kennedy](#), [Pierre Leonard](#), [Junhua Li](#), [Kristoffer Burgdorf](#), [Niels Grarup](#), [Torben Jørgensen](#), [Ivan Brandslund](#), [Henrik Bjørn Nielsen](#), [Agnieszka S. Juncker](#), [Marcelo Bertalan](#), [Florence Levenez](#), [Nicolas Pons](#), [Simon Rasmussen](#), [Shinichi Sunagawa](#), [Julien Tap](#), [MetaHIT consortium](#), ... [Oluf Pedersen](#)  [+ Show authors](#)

[Nature](#) 500, 541–546 (2013) | [Cite this article](#)

[10.1038/nature12506](https://doi.org/10.1038/nature12506)

- Individuals with a lower bacterial gene richness were characterized by more marked overall adiposity, insulin resistance and dyslipidaemia and a more pronounced inflammatory phenotype
- Individuals with lower number of genes (gene richness) and higher BMI had less efficient weight loss when following a diet

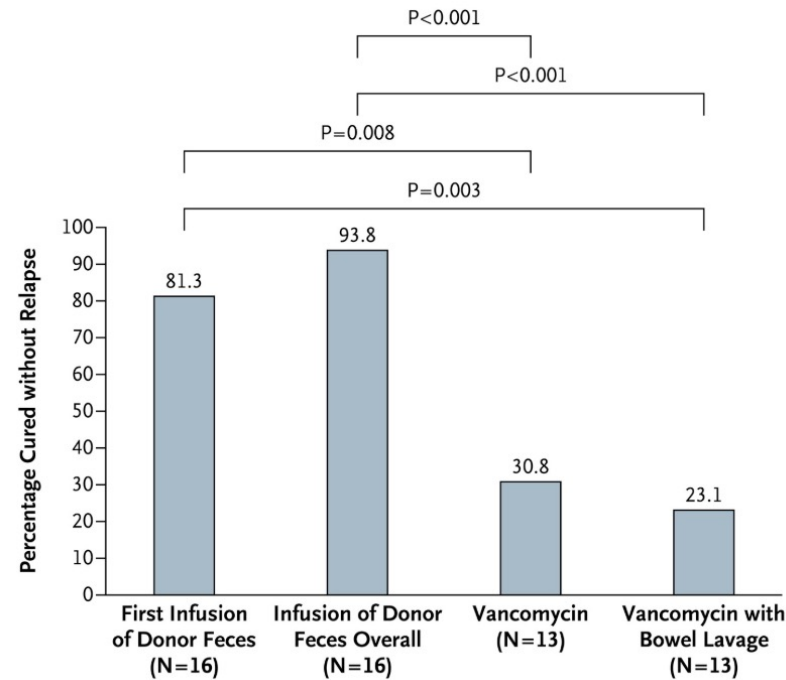
Figure 1: Distribution of low and high gene count individuals ($n = 292$).



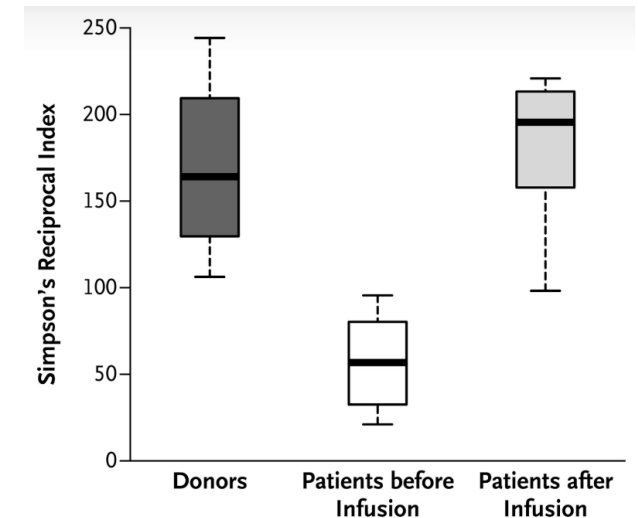
Fecal material transplantation (FMT) as experimental cure for *Clostridioides difficile*



[10.1056/NEJMoa1205037](https://doi.org/10.1056/NEJMoa1205037)

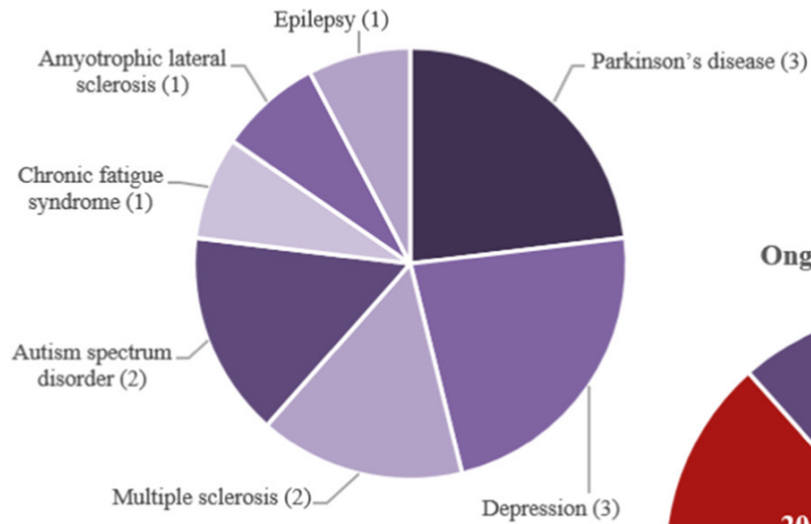


FMT successfully used to cure *Clostridioides difficile* infection with higher percentages of cured patients without relapse compared to the standard treatment (vancomycin)

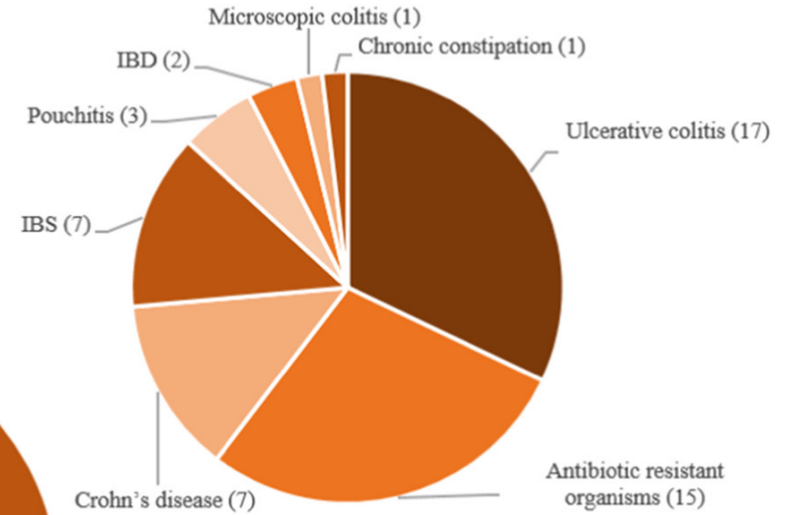


Diversity (Simpson's index) of patients after FMT resemble more the donors diversity

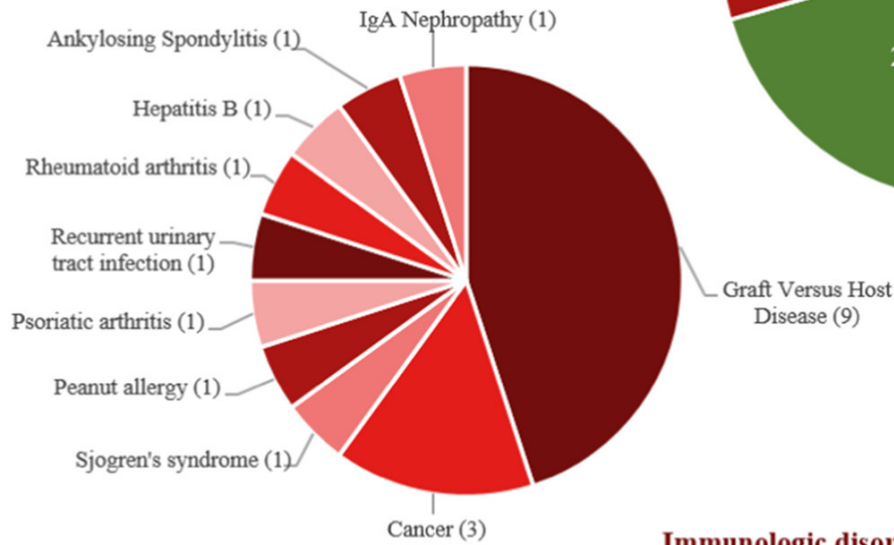
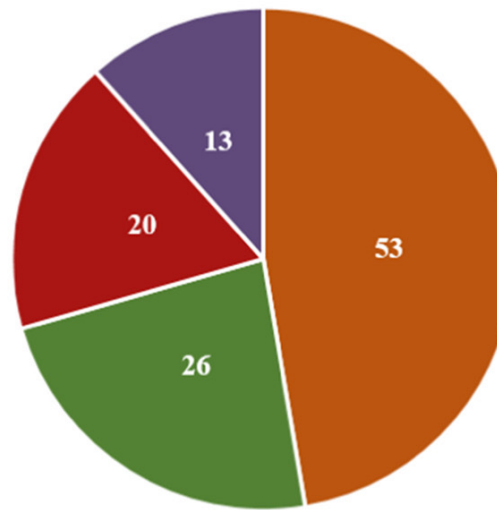
Neuropsychiatric disorders



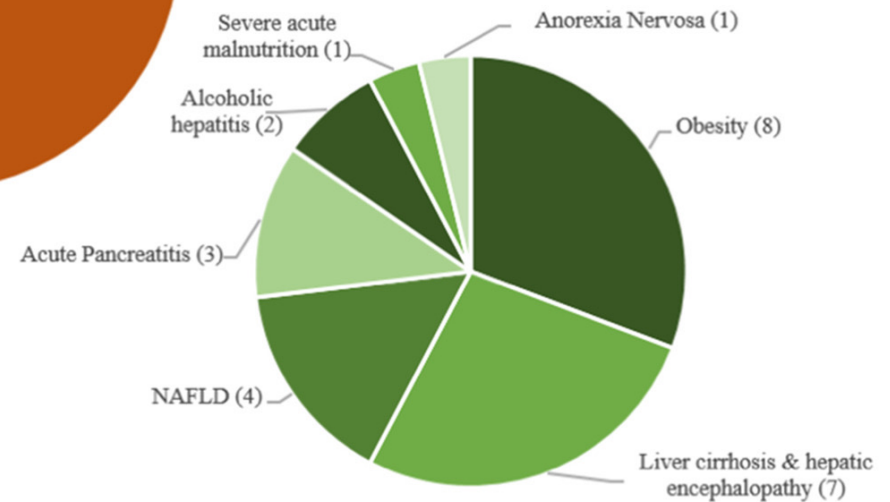
Gastrointestinal disorders



Ongoing trials with FMT



Immunologic disorder



Metabolic disorders

Resident gut bacteria can affect patient responses to cancer immunotherapy

REPORT



Gut microbiome influences efficacy of PD-1–based immunotherapy against epithelial tumors

BERTRAND ROUTY , EMMANUELLE LE CHATELIER , LISA DEROSA , CONNIE P. M. DUONG , MARYAM TIDJANI ALOU , ROMAIN DAILLÈRE , AURÉLIE FLUCKIGER , MERIEM MESSAOUDENE , CONRAD RAUBER, [...] AND LAURENCE ZITVOGEL +38 authors [Authors Info & Affiliations](#)

SCIENCE • 2 Nov 2017 • Vol 359, Issue 6371 • pp. 91–97 • DOI: 10.1126/science.aan3706 [DOI: 10.1126/science.aan3706](https://doi.org/10.1126/science.aan3706)

REPORT



The commensal microbiome is associated with anti–PD-1 efficacy in metastatic melanoma patients

VYARA MATSON , JESSICA FESSLER , RIYUE BAO , TARA CHONGSUWAT , YUANYUAN ZHA , MARIA-LUISA ALEGRE , JASON J. LUKE , AND THOMAS F. GAJEWSKI [Authors Info & Affiliations](#)

SCIENCE • 5 Jan 2018 • Vol 359, Issue 6371 • pp. 104–108 • DOI: 10.1126/science.aao3290 [DOI: 10.1126/science.aao3290](https://doi.org/10.1126/science.aao3290)

REPORT



Gut microbiome modulates response to anti–PD-1 immunotherapy in melanoma patients

V. GOPALAKRISHNAN , C. N. SPENCER , L. NEZI , A. REUBEN , M. C. ANDREWS , T. V. KARPINETS , P. A. PRIETO, D. VICENTE , K. HOFFMAN , [...] AND J. A. WARGO +60 authors [Authors Info & Affiliations](#)

SCIENCE • 2 Nov 2017 • Vol 359, Issue 6371 • pp. 97–103 • DOI: 10.1126/science.aan4236 [DOI: 10.1126/science.aan4236](https://doi.org/10.1126/science.aan4236)

In a series of back-to-back papers in Science:

- Routy et al. showed that higher abx consumption --> poorer response to cancer immunotherapy (immunotherapeutic PD1 blockade)
- Observed low levels of *Akkermansia muciniphila* in lung and kidney cancer patients → oral supplementation of this bacteria to abx-treated mice restored response to immunotherapy
- Matson et al. and Gopalakrishnan et al. studied melanoma patients receiving PD-1 blockade and found a greater abundance of “good” bacteria in the guts of responding patients.
- Non-responders had an imbalance in gut microbiota composition, which correlated with impaired immune cell activity.

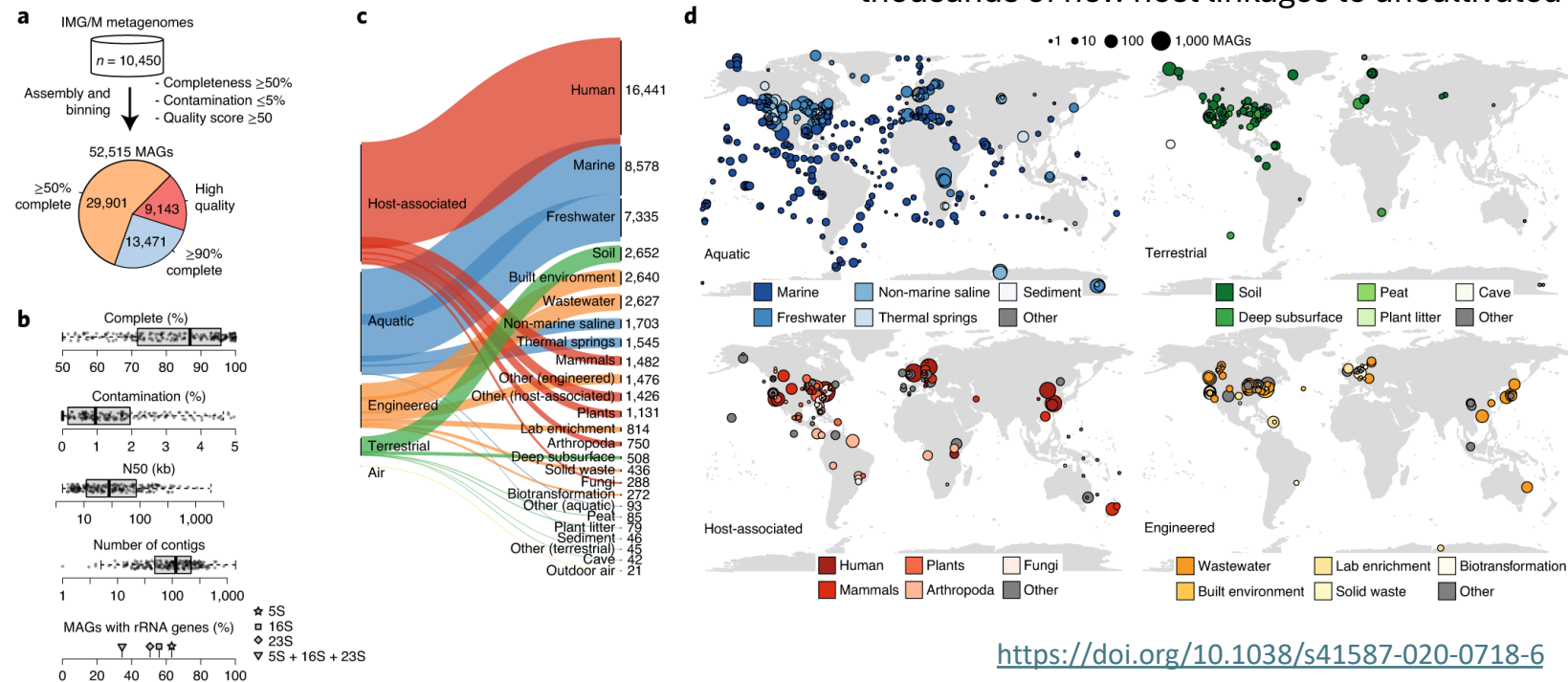
A genomic catalog of Earth's microbiome

A genomic catalog of Earth's microbiomes

Stephen Nayfach, Simon Roux, Rekha Seshadri, Daniel Udwarý, Neha Varghese, Frederik Schulz, Dongying Wu, David Paez-Espino, I-Min Chen, Marcel Huntemann, Krishna Palaniappan, Joshua Ladau, Supratim Mukherjee, T. B. K. Reddy, Torben Nielsen, Edward Kirton, José P. Faria, Janaka N. Edirisinghe, Christopher S. Henry, Sean P. Jungbluth, Dylan Chivian, Paramvir Dehal, Elisha M. Wood-Charlson, Adam P. Arkin, IMG/M Data Consortium, ... Emile A. Elie-Fadrosch ✉ [+ Show authors](#)

Nature Biotechnology 39, 499–509 (2021) | [Cite this article](#)

- >10,000 metagenomes collected from diverse habitats comprising host associated, aquatic, terrestrials, air and engineered microbiomes
- 52,515 metagenome-assembled genomes representing 12,556 novel candidate species from 135 different phyla
- The catalog allows metabolic modelling, understanding secondary-metabolite biosynthetic potential and resolving thousands of new host linkages to uncultivated viruses



<https://doi.org/10.1038/s41587-020-0718-6>

Antimicrobial resistance is a major threat to global health!

Genomic analysis of sewage from 101 countries reveals global landscape of antimicrobial resistance

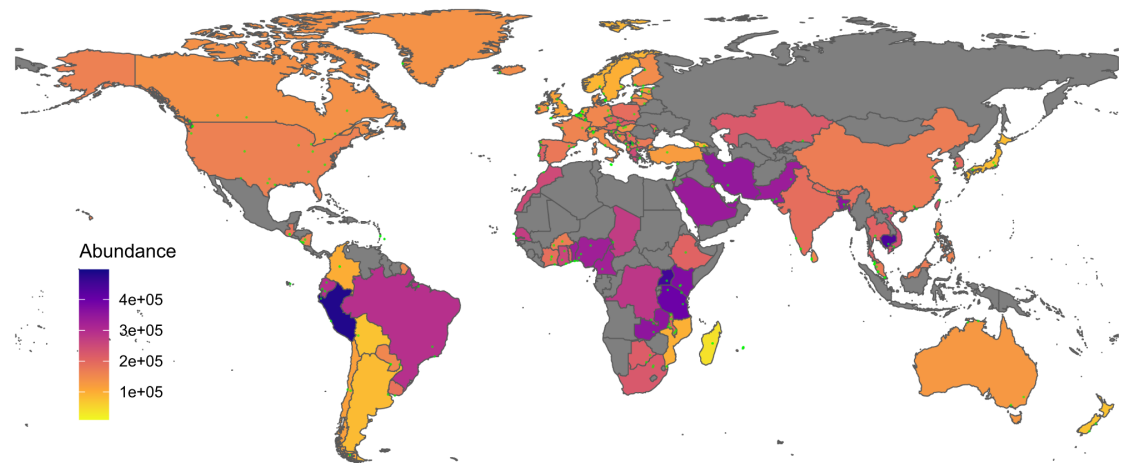
Patrick Munk , Christian Brinch, Frederik Duus Møller, Thomas N. Petersen, Rene S. Hendriksen, Anne Mette Seyfarth, Jette S. Kjeldgaard, Christina Aaby Svendsen, Bram van Bunnik, Fanny Berglund, Global Sewage Surveillance Consortium, D. G. Joakim Larsson, Marion Koopmans, Mark Woolhouse & Frank M. Aarestrup

Nature Communications 13, Article number: 7251 (2022) | [Cite this article](#)

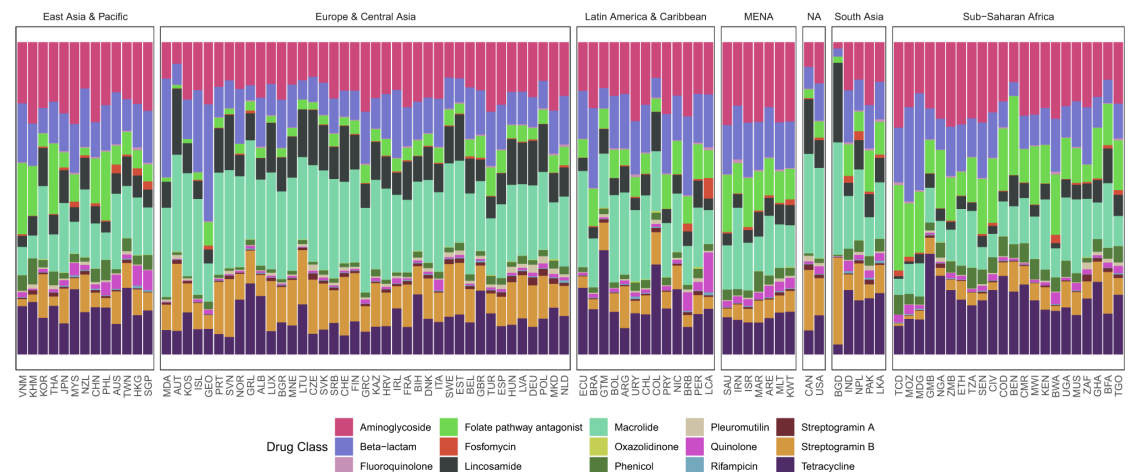
<https://doi.org/10.1038/s41467-022-34312-7>

- The global resistome based on sewage-based monitoring
- Metagenomic samples revealed regional patterns reflecting taxonomical changes
- Understanding the emergence, evolution, and transmission of individual antibiotic resistance genes (ARGs) is essential to develop sustainable strategies to fight this threat

a



b



Microbial community at the Deepwater Horizon oil spill site

JOURNAL ARTICLE

Metagenomics reveals sediment microbial community response to Deepwater Horizon oil spill

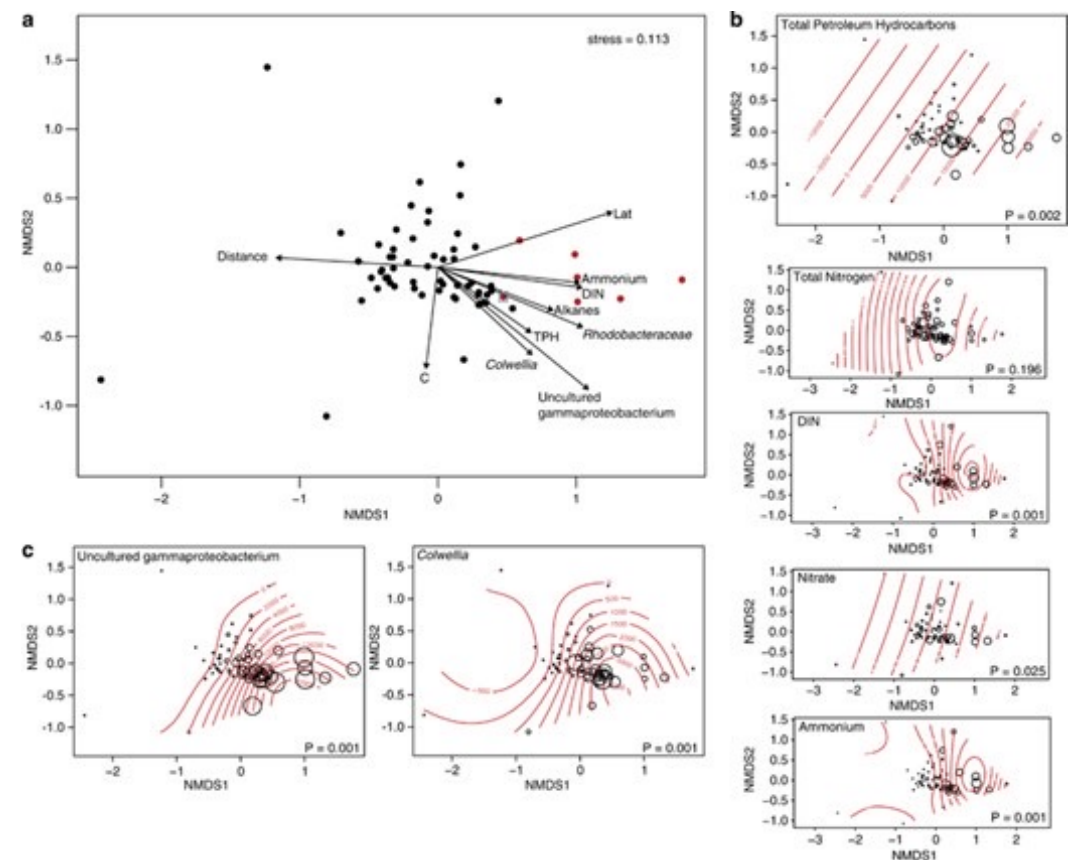


Olivia U Mason , Nicole M Scott, Antonio Gonzalez, Adam Robbins-Pianka, Jacob Bælum, Jeffrey Kimbrel, Nicholas J Bouskill, Emmanuel Prestat, Sharon Borglin, Dominique C Joyner ... [Show more](#)

The ISME Journal, Volume 8, Issue 7, July 2014, Pages 1464–1475,

<https://doi.org/10.1038/ismej.2013.254>

- Metagenomic Analysis of sediment samples from 64 sites in the Gulf of Mexico, characterizing microbial community's response to the oil spill
- Uncultured *Gammaproteobacterium* and *Colwellia* species were enriched in oil-impacted sediments (hydrocarbon degradation)
- An increase in genes associated with hydrocarbon degradation pathways (aliphatic and aromatic hydrocarbon breakdown), indicating adaptive microbial response to oil contamination
- The study emphasized the importance of microbial communities in natural bioremediation processes.



Global ocean microbiome (Tara oceans)

Article | [Open access](#) | Published: 11 June 2018

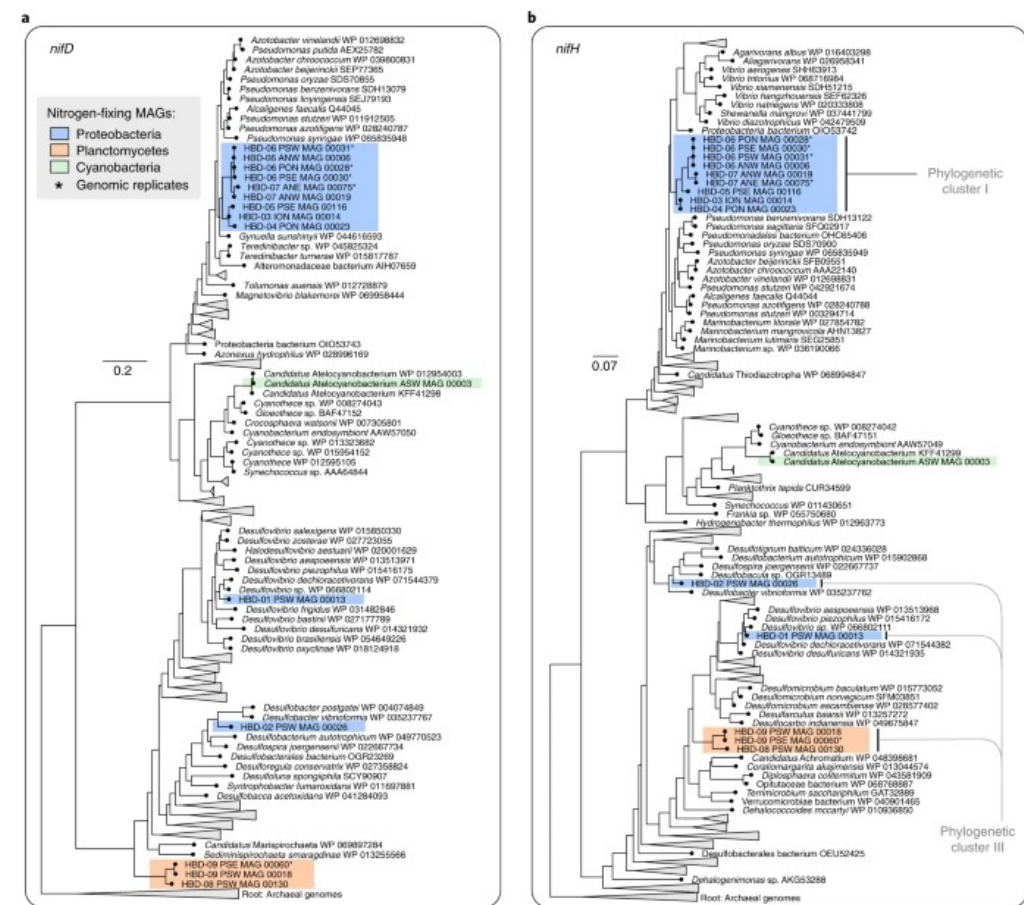
Nitrogen-fixing populations of Planctomycetes and Proteobacteria are abundant in surface ocean metagenomes

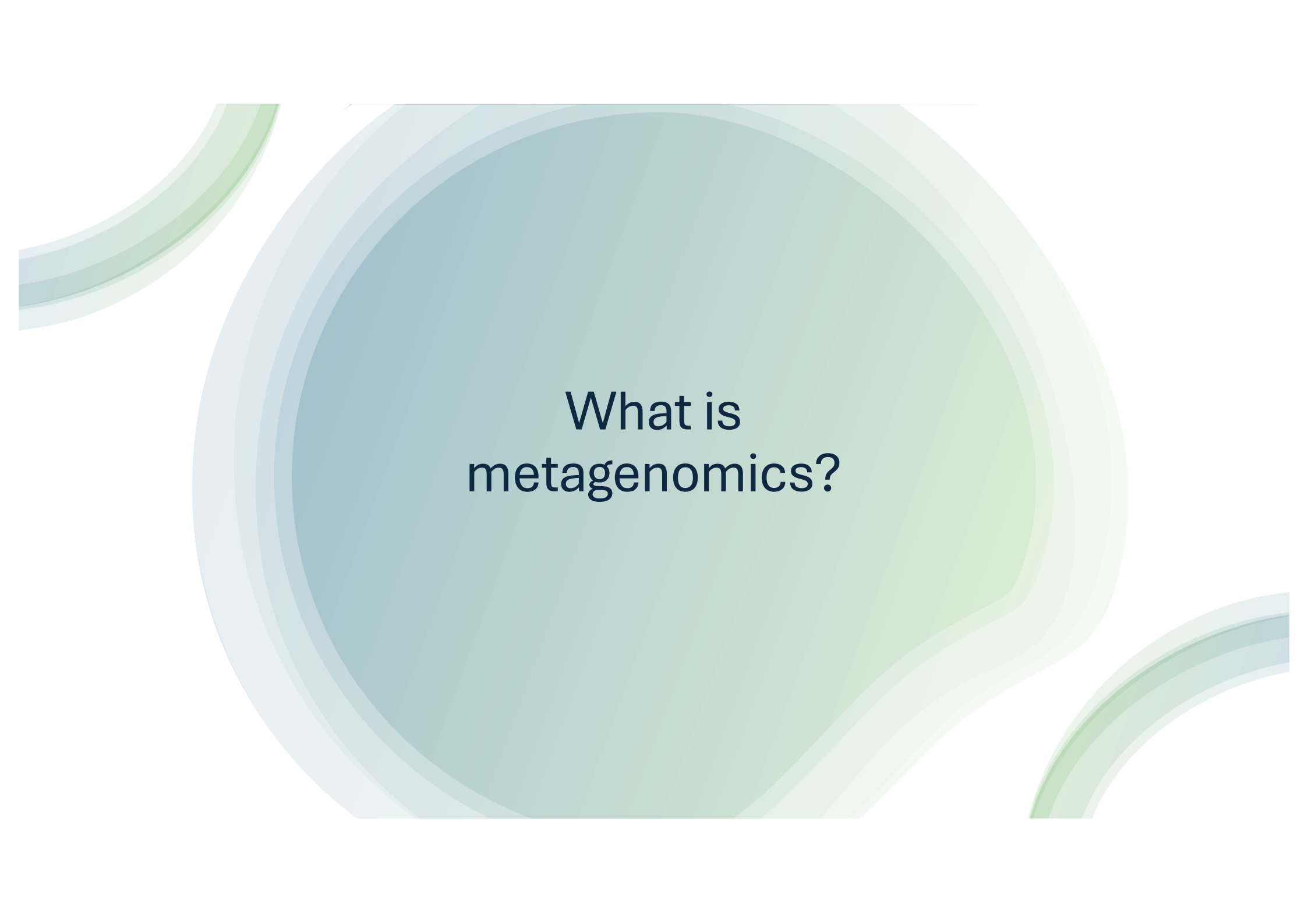
[Tom O. Delmont](#) , [Christopher Quince](#), [Alon Shaiber](#), [Özcan C. Esen](#), [Sonny TM Lee](#), [Michael S. Rappé](#), [Sandra L. McLellan](#), [Sebastian Lückert](#) & [A. Murat Eren](#) 

Nature Microbiology 3, 804–813 (2018) | [Cite this article](#)

<https://doi.org/10.1038/s41564-018-0176-9>

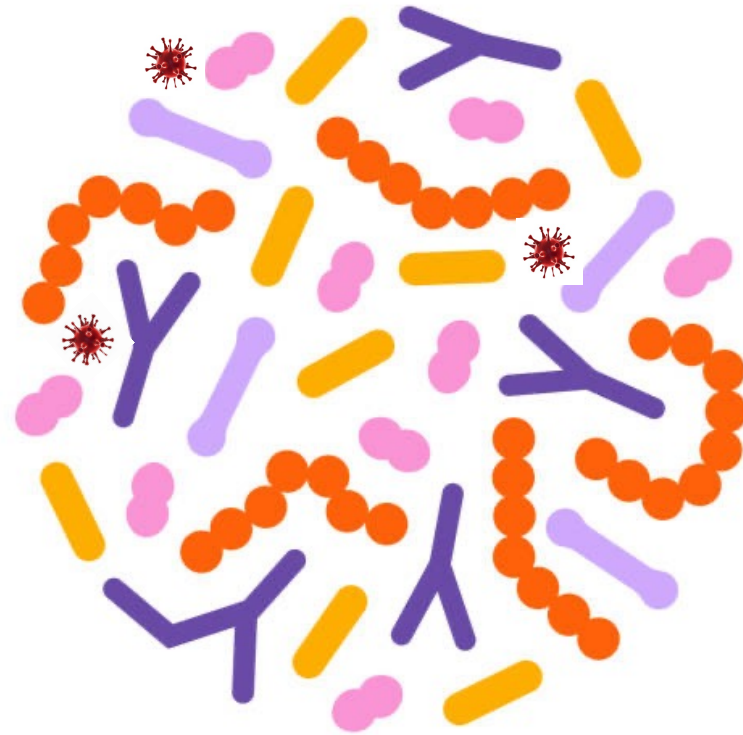
- Characterized ~1,000 non-redundant microbial genomes from the TARA Oceans sites
- Nitrogen fixation in the surface ocean impacts global marine nitrogen bioavailability and therefore the microbial primary productivity
- Usually done by cyanobacteria, but non-cyanobacterial diazotrophs inhabiting surface waters of the open ocean (lineages within the Proteobacteria and Planctomycetes) have also the functional potential (nifD, nifH)





What is
metagenomics?

Microbiome definition



Microbiome definition

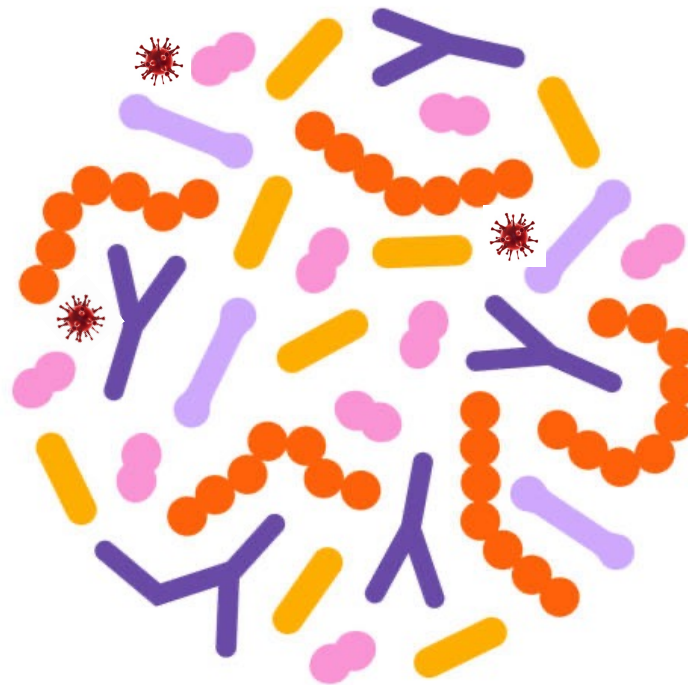
Microbiome refers to an microbial community in an specific environment

Inhabitants:

- Bacteria
- Archaea
- Fungi
- Viruses
- Protozoa
- ...

Environment:

- Soil
- Ocean
- Wastewater plant
- Gut
- Skin
- ...



Organism products:

- Genes
- Proteins
- Pathways
- Metabolites

In contrast to a monoculture (all cells belong to one organism), in a microbiome we have cells of many different organisms → microbial community

What a microbiome study usually answer?

What a microbiome study usually answer

Who is living in this habitat?

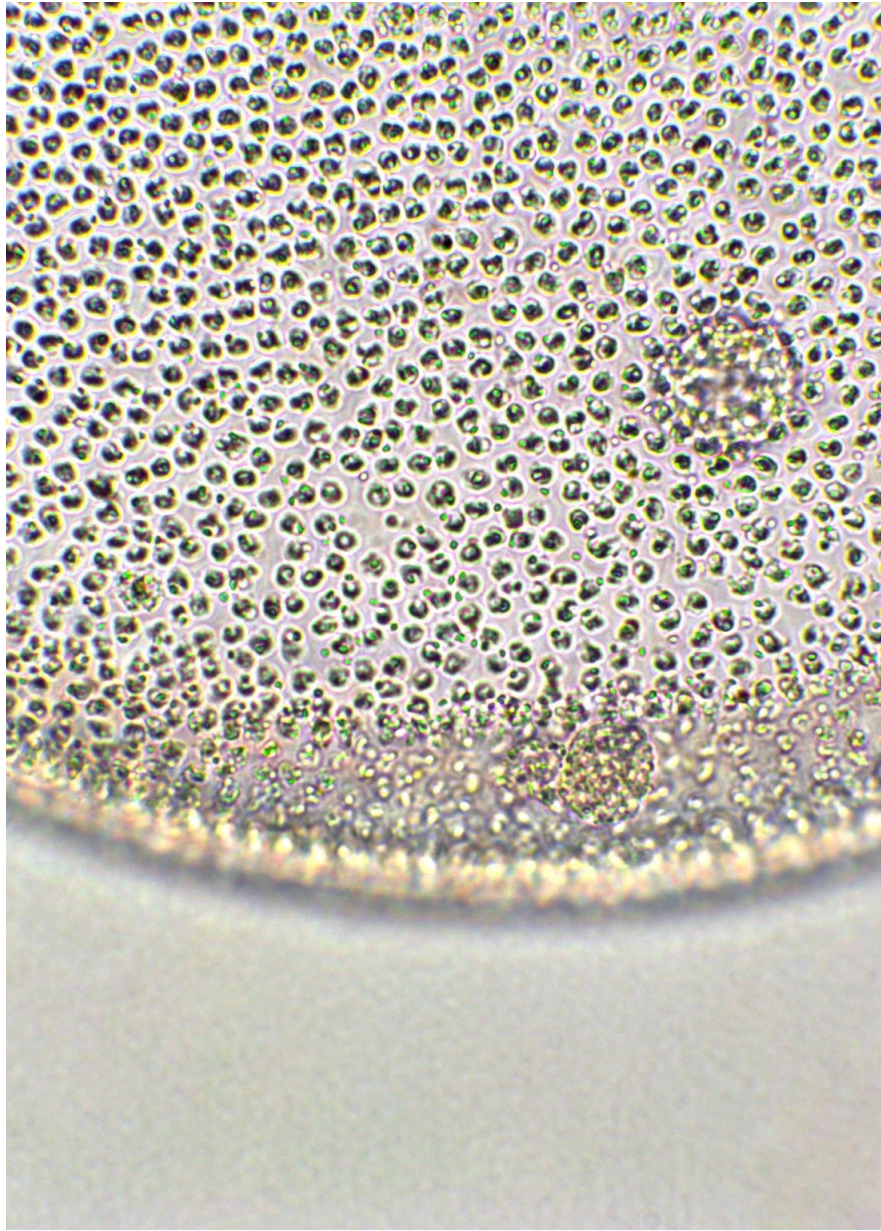
What is the relative proportion of the different members
What is the total abundance (number of cells) (by absolute abundance)
How abundant are bacteria, archaea, protozoa, virus, ...
How diverse is the microbial community (diversity indices)
How evenly distributed is it?

What are these organisms doing?

What functions potentially do they perform?
What substrates they consume?
What metabolites they produce?

How do they perform their activities?

Which genes and/or pathways are present?

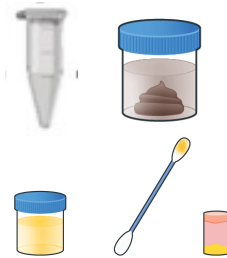
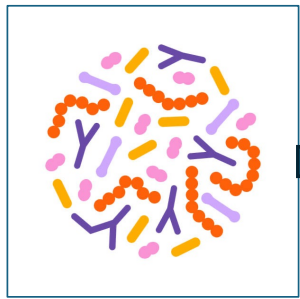


What is Metagenomics?

- Study of the genetic material sampled directly from the environment (fecal, skin, soil, marine,... samples)
- No need of culturing organisms – many of the organisms that we characterize can not even be cultured – we can study all the organisms in the microbial community
- Based on high-throughput DNA sequencing technology to analyze microbial diversity, taxonomic and functional profiles, and microbial interactions

Experimental design

DNA Sampling
spatula, swabs,
filtration units,
soil cores,...



DNA Storage
freezing,
preservation
buffers,...



DNA Extraction
Kits to lyse the
cells and release
the DNA



DNA Sequencing
library prep: fragmentation,
end repair, adapter ligation



Taxonomical and
functional
characterization of the
sequenced reads
(fastQ files)



Microbiome abundances

OTU	Sample 1	Sample 2	Sample 3
A	0.60	0	0.70
B	0.24	0.20	0.10
C	0.10	0	0
D	0.05	0	0
E	0.01	0	0
F	0	0.80	0.20
Total	1.0	1.0	1.0

**Data
Analysis**

Sample collection

- To obtain a **representative and uncontaminated snapshot of a microbial community** in a specific environment (e.g., gut, soil, water, skin,...)
- **Sterile tools** and containers need to be used
- Microbial DNA can degrade rapidly. A good preservation ensures the microbial community structure remains as close to in situ as possible.
 - **Freezing** (-20 to -80C) for long-term storage
 - Preservation buffers if freezing is delayed
- For sample transportation use **cold chain logistics** (ice packs, dry ice, liquid nitrogen)
- **Label** clearly with metadata: time, location, environmental conditions, etc.
 - Important for downstream analysis and reproducibility



DNA Extraction

- To obtain **high-quality, unbiased, and inhibitor-free DNA** from a complex matrix (e.g., soil, feces, water)
- Isolate **total microbial DNA** from all organisms in a sample (bacteria, archaea, fungi, viruses—depending on context)
- Steps:
 - **Cell Lysis:** The goal is to **lyse all types** of microbes, including tough ones like Gram-positive bacteria or spores.
 - Mechanical disruption (Bead beating)
 - Chemical/enzymatic lysis (detergent/enzymes (lysozyme, proteinase K, Rnase))

Kits are preferred in metagenomics for consistency, ease, and reduced contamination risk (e.g., Qiagen, Zymo, MoBio/DNeasy)

- **Contaminant Removal:** Use specialized kits with inhibitor removal steps (e.g., PowerSoil DNA Kit)
- **DNA Purification:** remove proteins, RNA, lipids, and inhibitors
- **Quantification & Quality Check**

Common DNA extraction kits:

- QIAGEN DNeasy PowerSoil Pro Kit
- QIAGEN QIAamp PowerFecal Pro DNA Kit
- ZymoBIOMICS DNA Miniprep Kit



Sequencing

- To read the **nucleotide sequences** of DNA fragments extracted from a microbial community — enabling identification of **who is present** and **what they can do**
- **Library preparation:** DNA must be **fragmented** and **tagged with adapters** for compatibility with sequencing platforms.
- Include negative and positive **controls** (e.g., mock communities)
- Choose **depth** wisely: deeper for complex samples or analysis (e.g., soil, strain resolution, single nucleotide variants(SNVs)), shallower for simpler ones
- Paired-end sequencing from Illumina (Sequencing by Synthesis):
 - <https://www.youtube.com/watch?v=HMyCgWhwB8E>



Platforms

Platform	Features	Read Length	Use Case
Illumina	High accuracy, short reads	~150–300 bp	Most common for shotgun & amplicon
Oxford Nanopore	Long reads, portable	up to 100 kb+	Assembly, rare taxa detection
PacBio HiFi	Long + accurate reads	~10–25 kb	Functional profiling, assembly
Ion Torrent	Medium throughput	~200–400 bp	Less common but used in clinical



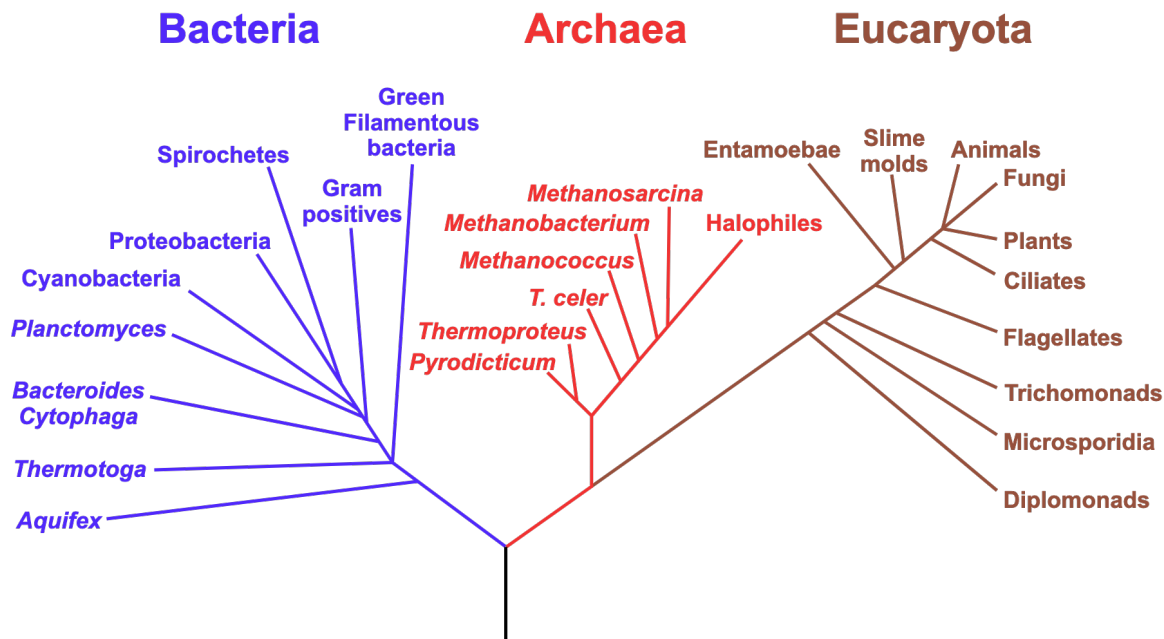
Taxonomical profiling

Taxonomical profiling

- Process of identifying and quantifying the microbial taxa present in a sample - this can be bacteria, archaea, viruses, fungi, ...
- Typically from metagenomic or amplicon sequencing data (metabarcoding).
- The most common taxonomical profiling methods can be grouped based on the type of sequencing data used:
 - Amplicon based profiling: 16S, 18S, ITS rRNA sequencing – e.g. 16S rRNA gene is used to profile bacteria and archaea
 - Whole-genome shotgun (WGS) metagenomics involves random sequencing of all genetic material in a sample:
 - Reference-based methods: direct mapping of reads to genes, proteins or genomes
 - Assembly-based methods: assemble reads into contigs and bin them into metagenome/assembled genomes

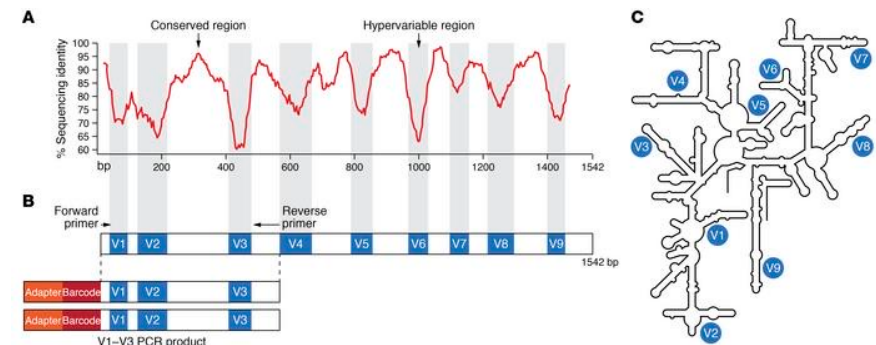
16S amplicon sequencing

Phylogenetic Tree of Life



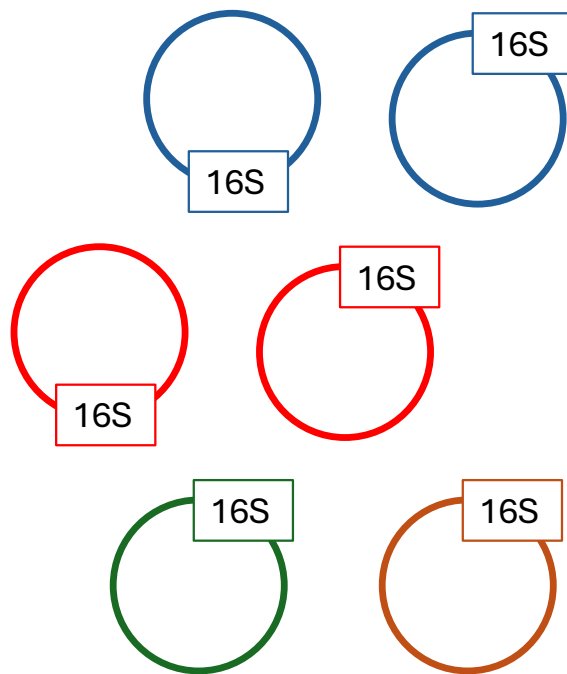
Universal phylogenetic tree based on the 16S rRNA gene sequence comparisons.
 Pace, N. A molecular view of microbial diversity and the biosphere.
 Science 276:734-740. 1997.

- 16S rRNA gene is an ancient gene that encodes the RNA component of the small (30S) subunit of the prokaryotic ribosome
- As it is conserved in many organisms, traditionally used to infer phylogenetic trees to figure out species evolution (Present also in mitochondria and chloroplasts of eukaryotes)
- 16S gene has
 - conserved regions: spread around all organisms and easy to amplify
 - variable unique regions: to tag specifically species

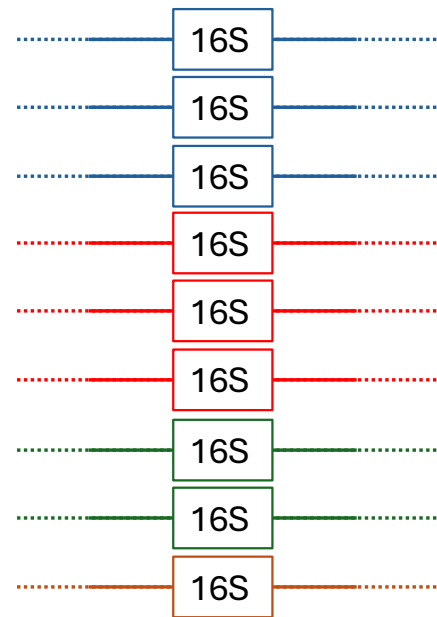


J Clin Invest. 2022;[132\(7\)](https://doi.org/10.1172/JCI154944):e154944.
<https://doi.org/10.1172/JCI154944>.

16S amplicon sequencing



Microbial genomes in our sample carrying 16S rRNA gene



PCR amplification of 16S rRNA gene to produce a pool of DNA containing only 16S genes

Sequencing all fragments

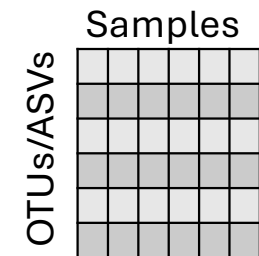
TT**C**AACCGAGATATGC
 TT**C**AACCGAGATATGC
 TT**C**AACCGAGATATGC
 TTTAACC**G**AT**T**ATATGC
 TTTAACC**G**AT**T**ATATGC
 TTTAACC**G**AT**T**ATATGC
 TTTAAC**A**GAGATATGC
 TTTAAC**A**GAGATATGC
 TTTAACCGAGATAT**A**C

Sequencing all fragments

Traditionally cluster the sequences (97% similarity) into operational taxonomic units (OTUs) and compare to known 16S databases



assign taxonomy directly to the denoised amplicon sequence variants therefore allowing more precise taxonomic placement than OTUs



Abundance table of operational taxonomic units (OTUs) or amplicon sequence variants (ASVs) – resolution up to Genus and no functional annotation

A diagram consisting of six circles arranged in two vertical columns of three. The left column contains a blue circle at the top, a red circle in the middle, and a green circle at the bottom. The right column contains a blue circle at the top, a red circle in the middle, and a brown circle at the bottom. The circles are of varying sizes and are not connected to each other.

TTCAACCGAGATATGC
AGACGCAGATATGC
GAACGGAGAAATCC
ACAACCGAGATATGC
TTCAACCGAGATATGC
ATCAACGGAGATATGC
TTCAACCGAGATATGC
TACAACCGAGATGC
ATCAACACCGAGATATGC
ATCAACACATATGC
TTCTTCGGAGATTTC
GACTCGAGTTTTCC
GAACGGAGATATGC
AGACGGAGATATGC

Read mapping to know genomes, specific marker genes (BLAST, bowtie2, DIAMOND)

----- AACACCGAGATATCC ----- TTCAACGGAGATATGC -----
AACACCGAGATATCC TTCAACGGAGATATGC

TTCAACCGAGATATGC
TTCAACCGAGATATGC
TTCAACCGAGATATGC
...

ATCAACA
ATCAACA
...

ATCTTCA

[illegible]

Abundance table

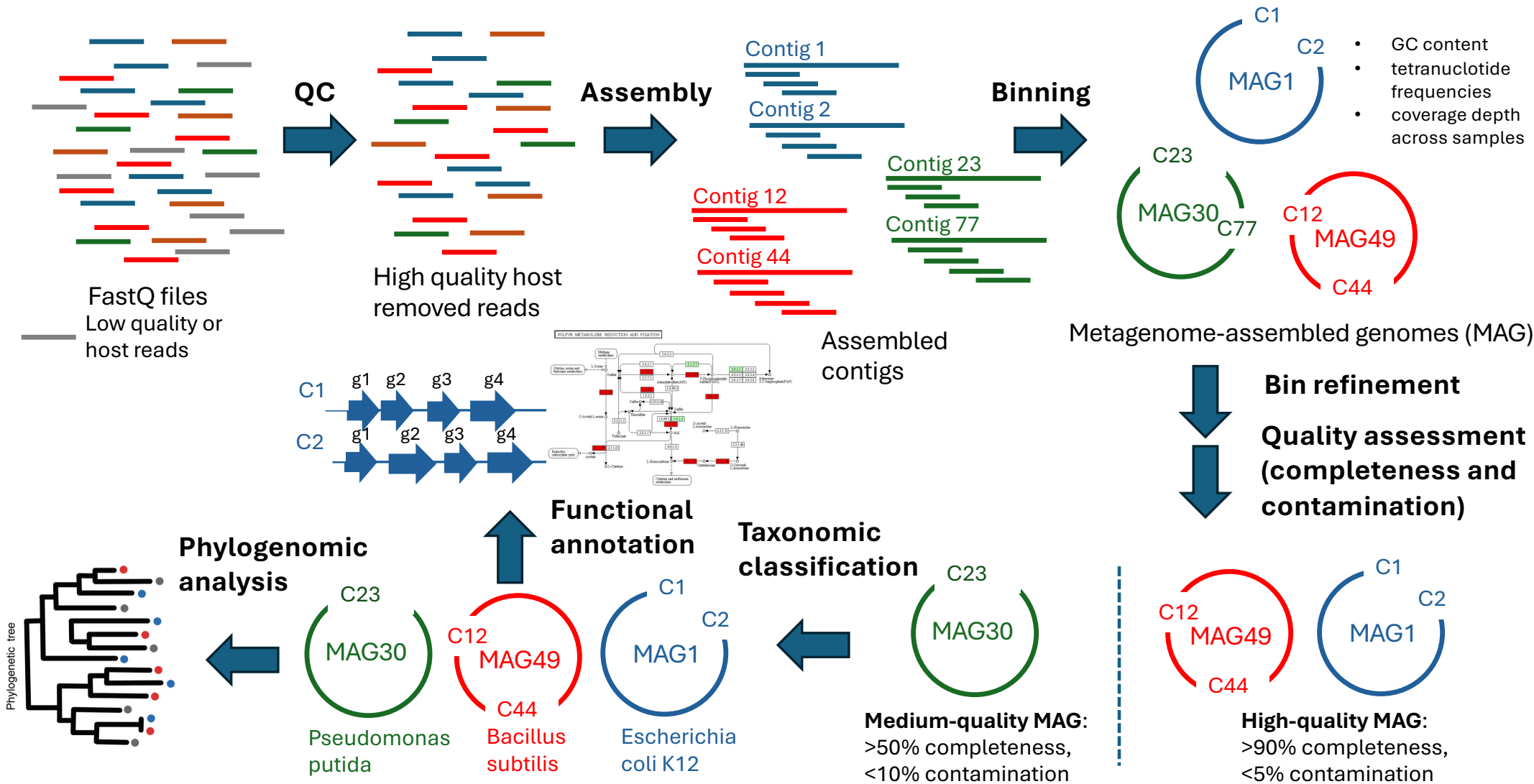
Amplicon and read-based shotgun metagenomics tools

Method	Tools	Databases
16S Amplicon-based Profiling	QIIME 2 : A popular pipeline supporting DADA2, Deblur, and multiple classifiers (Naive Bayes, etc.). DADA2 : High-resolution method that infers amplicon sequence variants (ASVs). USEARCH / VSEARCH : Clustering-based OTU picking methods. Mothur : OTU-based method using different clustering and alignment strategies. RDP Classifier : Naive Bayesian classifier for assigning taxonomy.	SILVA Greengenes (deprecated but still used) RDP UNITE (for fungi)
Shotgun Metagenomics-based Reference-based profiling	MetaPhlAn4 : Uses clade-specific marker genes for fast, accurate profiling. Kraken2 / KrakenUniq : K-mer based, ultra-fast classification against a database of genomes. Centrifuge : K-mer based, designed for large reference databases and low-memory use. HUMAnN / HUMAnN3 : Often used alongside MetaPhlAn for functional and taxonomic profiling. StrainPhlAn : Extension of MetaPhlAn for strain-level profiling.	ChocoPhlAn (used by MetaPhlAn) RefSeq / GenBank (used by Kraken/Centrifuge) GTDB (Genome-based taxonomic reference)

Choice depending on:

- **Speed vs. Accuracy**: Kraken2 is fast; MetaPhlAn is more accurate for species-level identification.
- **Resolution**: DADA2 and MetaPhlAn can provide high-resolution profiling.
- **Data Type**: Amplicon (16S) vs. shotgun metagenomics.
- **Resource Requirements**: Kraken needs large memory for big databases; MetaPhlAn is lighter.

Assembly-based methods: Metagenome-assembled genomes (MAG)



MAG construction tools

<https://pubmed.ncbi.nlm.nih.gov/28898207/>

Step	Goal	Tools	Notes
Quality Control	Remove low-quality reads, adapters, and contaminants (e.g. host reads)	fastp, adapterremoval, Trimmomatic, BBduk	Output: Clean reads
Assembly	Assemble short reads into longer contigs.	MEGAHIT, metaSPAdes	Output: Assembled contigs
Binning	Group contigs that belong to the same genome	MetaBAT2, MaxBin2, CONCOCT, VAMB	Output: bins of contigs belonging to the same genome Using coverage profiles and sequence composition (GC content, tetranucleotide frequency).
Bin refinement	Improve bin quality and merge results from different binning tools	DASTool, metaWRAP	Output: Refined bins
Quality Assessment	Assess completeness and contamination of bins (MAGs)	CheckM, BUSCO	Metrics usually used: High-quality MAG: >90% completeness, <5% contamination. Medium-quality MAG: >50% completeness, <10% contamination.
Taxonomic classification	Assign taxonomy to MAGs	GTDB-Tk, Kraken2, CAT/BAT	Output: taxonomically annotated MAGs
Functional annotation	Predict genes and metabolic functions	Prokka, EggNOG-mapper	Out put: functionally annotated MAGs
Phylogenomic analysis	Compare MAGs evolutionarily	PhyloPhlAn, Anvi'o, IQ-TREE	Output: Phylogenetic trees based on MAGs

Comparing the three methods

	Amplicon/16S sequencing	Shotgun metagenomics	
		Reference - based	Assembly - based
Pros	• Can handle low biomass / non-bacterial contamination • Rough taxonomy of novel species • Less expensive	• High resolution (subspecies, genes SNVs) • Can detect all DNA in the sample (also virus, fungus, protozoa,...) • Fast and scales well • Finds rare entities	• High resolution (subspecies, genes SNVs) • Can detect all DNA in the sample (also virus, fungus, protozoa,...) • Can identify novel genomes, genes and synteny patterns
Cons	• Low resolution • Limited to bacteria or archaea • No subspecies or strains resolution • Biases due to PCR amplification / amplicon copy number variation	• Does not work well for low biomass • More expensive than 16S • Limited to what is known in the databases	• Does not work well for low biomass • More expensive than 16S • Misses rare entities • Computationally very intensive



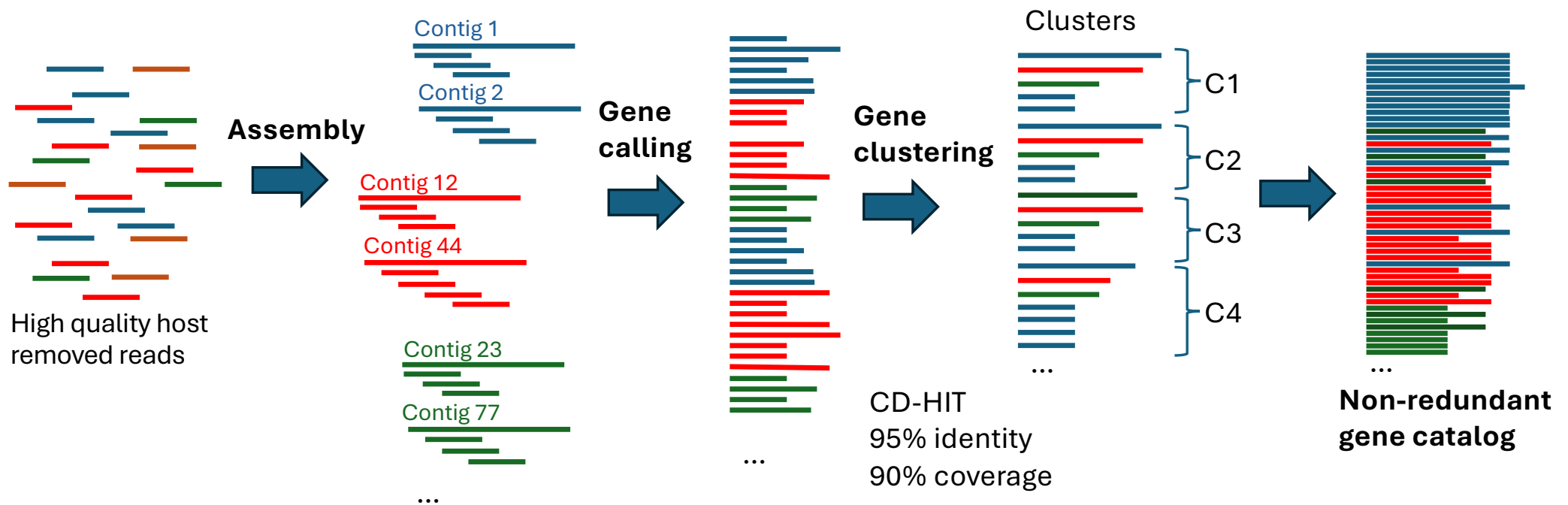
Functional profiling

Building a microbial gene catalog

To do functional annotation you can always map reads to genes (read/reference-based annotation)

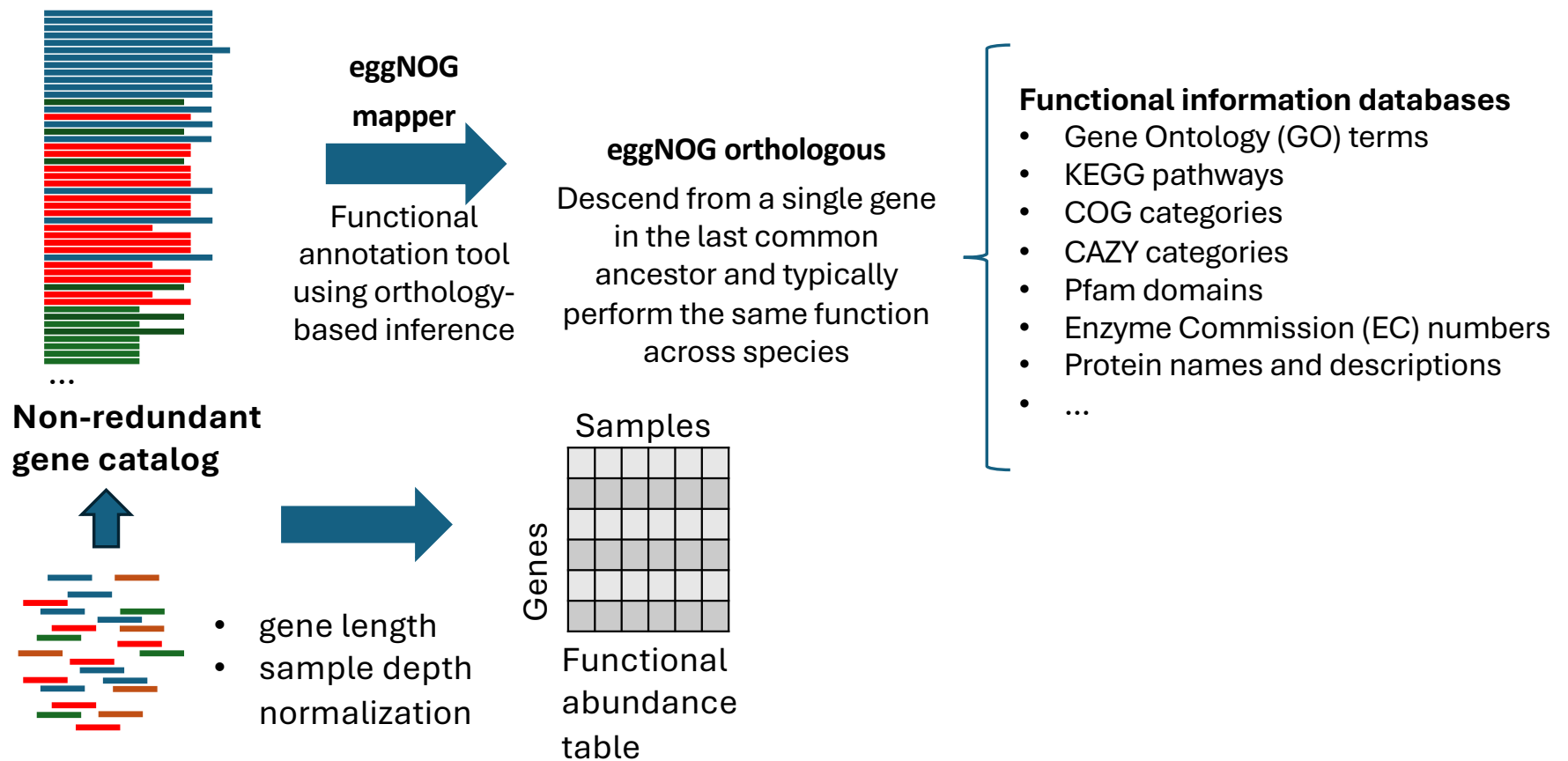
A common resource used in functional annotation is building a gene catalog

The beginning of the process is the same as the assembly method



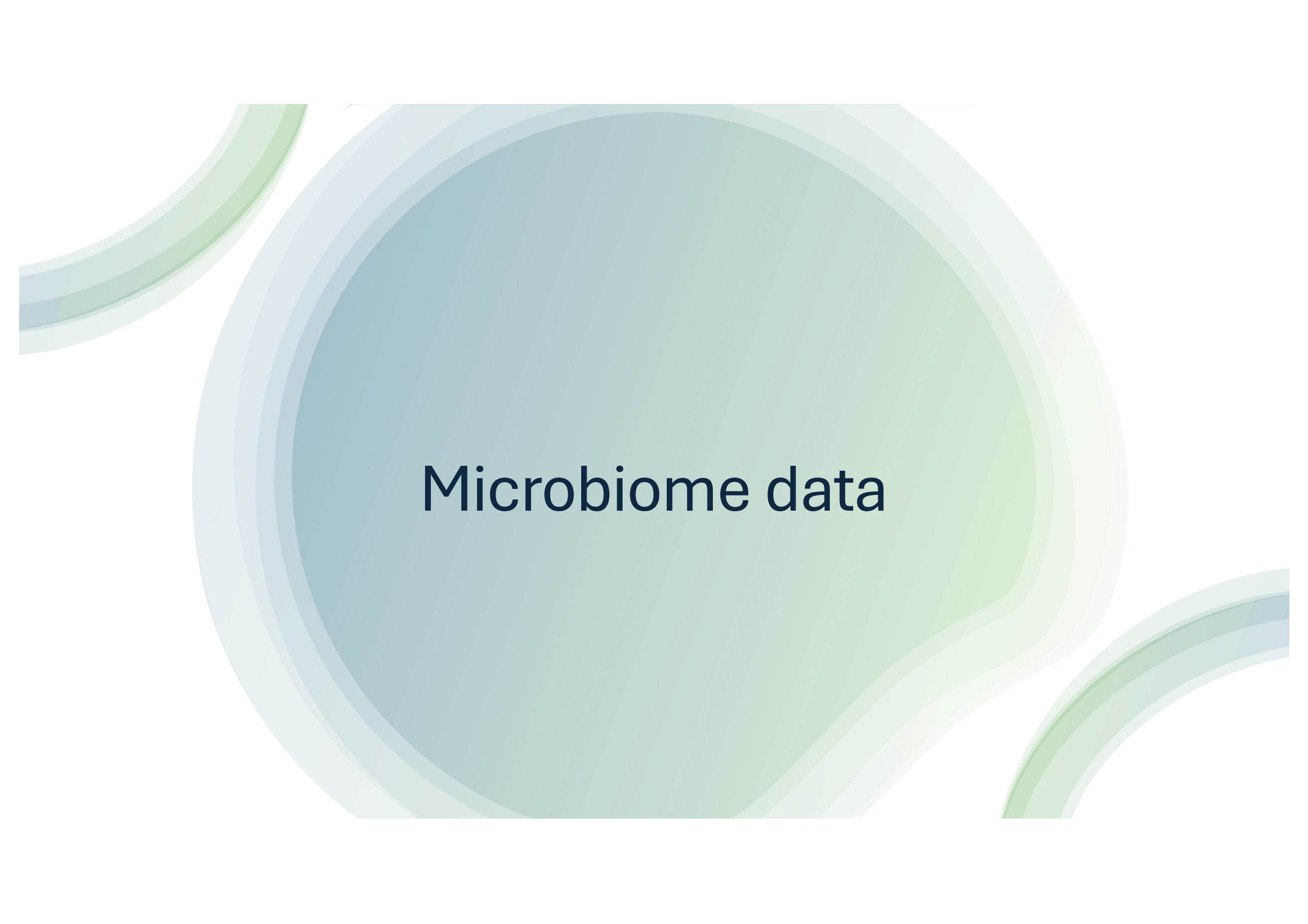
Annotating the gene catalog and getting functional abundance

Reads can be mapped to the catalog genes to infer gene abundance and those genes can be mapped to several functional information databases (e.g KEGG database, PATRIC Database, CARD or Resfam's antibiotic resistance gene databases,...)



Why building a gene catalog?

- Annotation of genes across all metagenome (not only the known ones – novel or uncharacterized functions - new biochemical functions, pathways, or resistance mechanisms)
- Cross-sample comparability - All samples are quantified against the same gene set, then cleaner statistics and fewer biases.
- Computational efficiency (downstream) - Annotation and indexing are done once instead of per sample.
- Better detection of shared vs unique genes - You can easily identify core vs accessory genes across samples.
- Reduce redundancy - Many microbes share similar genes. Clustering genes removes duplicates and simplifies downstream analysis
- Reusable resource - valuable resource for upcoming large-scale projects



Microbiome data

Microbiome data

- **Sparsity**
- **Dynamic range**
- Usually expressed as Relative, not Absolute abundance
- Compositional data

		Relative abundance (%)								
Area		1	2	3	1	2	3	1	2	3
Condition		X	X	X	Y	Y	Y	Z	Z	Z
Relative abundance Taxa or functional feature	10	10	11	20	12	14	16	25	34	22
	70	70	49	56	0.9	0.5	0	0	0.1	0
	0	0	0	0	0.4	0.1	0	0	0	0
	0.3	0.3	0.4	0.45	0	0.1	1	2	3	4
	2	2	2	3	0	0	0.01	1	1	2
		Samples								

Microbiome data

Microbiome data is compositional:

- Sparsity
 - Dinamic range
 - **Usually expressed as Relative, not Absolute abundance**
 - **Compositional data**
- Sampling of a fraction of the total genomic DNA
 - Different sequencing depth between samples
 - Both sampling depth and sequencing depth can vary several orders of magnitude between samples
 - Absolute abundances are normally not known – abundances are relative and constraint to sum to 1 (or 100%)
 - Proportions or percentages of total reads per sample — not true cell counts.
 - Relative abundances are susceptible to compositional effects

<https://pubmed.ncbi.nlm.nih.gov/27306058/>

Compositional data

Consider read counts as a proxy for cell abundance (not perfect as 16S gene can have multiple gene copies in some genomes, and in shotgun metagenomics we have different genome sizes – larger genomes contributing with more reads per cell, extraction efficiency, library prep bias, etc ...)

	S1 cells	S1 Rel ab %	S2 cells	S2 Rel ab %
Taxa 1	250	25	250	33.33
Taxa 2	250	25	250	33.33
Taxa 3	300	30	50	6.67
Taxa 4	200	20	200	26.67
Total	1000	100	750	100

- Only Taxon 3 reduced its abundance after antibiotic treatment, but 1,2, and 4 seem that they increase
- Affects downstream analysis and interpretations
 - e.g. correlations or regression assume independent variables
 - analysis alternatives and data transformations have been proposed: center-log-ratio transformation (clr), ALDEx2, ANCOM
 - however, they introduce practical limitations — especially with zero handling, computational cost, and difficult interpretation

Ideal scenario: absolute abundances

If we want to know **true cell counts**, we need to:

- Add **internal spike-in controls** (known amounts of reference DNA).
- Use **qPCR or flow cytometry** to estimate **total bacterial load**.
- This allows you to correct read counts to reflect **absolute abundance**, not just proportions.



Diversity measures

Diversity measures:

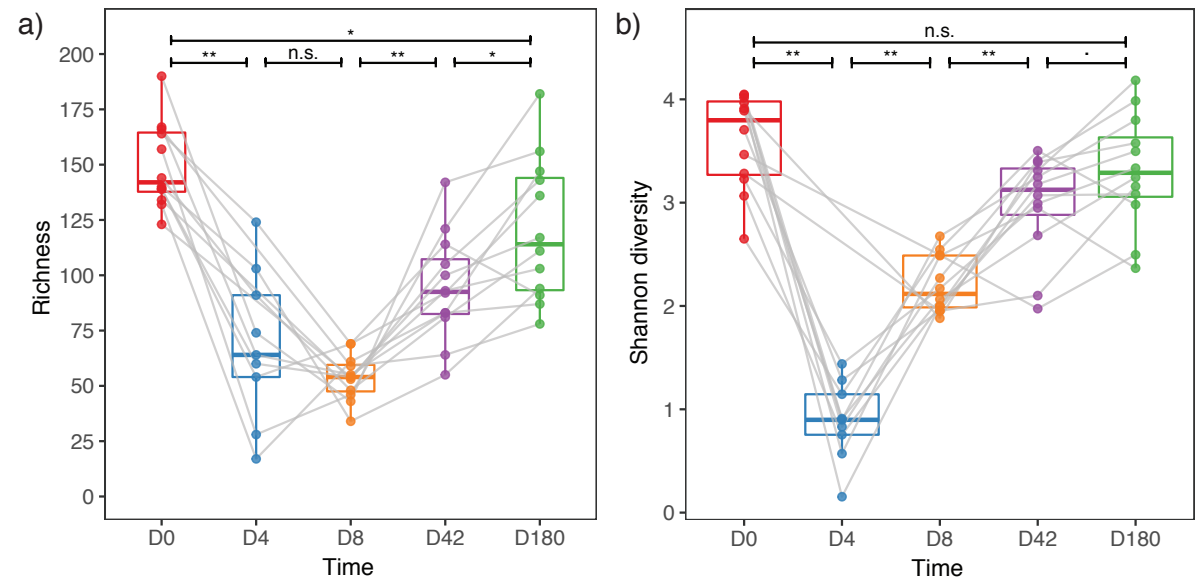
Alpha diversity

- Shannon index:

$$H' = - \sum_{i=1}^S p_i \cdot \ln(p_i)$$

- S = total number of species
- p_i = proportion of species i in the sample

- Diversity within a sample
 - Richness:** number of unique taxa found in a sample (species, OTUs, ASVs,...)
 - Shannon Index:** Accounts for both richness and evenness (how balanced the community is)
 - Higher values** = More diverse and evenly distributed community
 - Other indices: Simpson, Chao1, Faith's PD (for phylogenetic diversity)

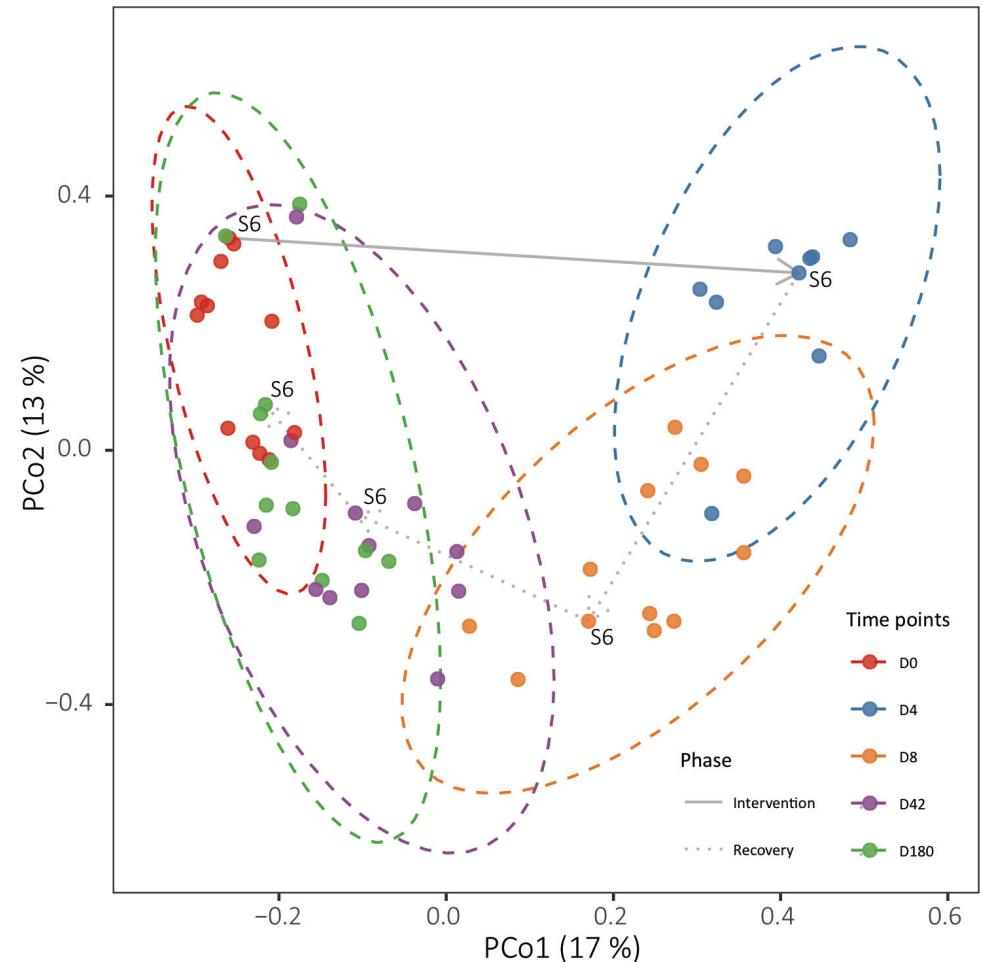


<https://pubmed.ncbi.nlm.nih.gov/30349083/>

Diversity measures:

Beta diversity

- Diversity between samples (community composition dissimilarity)
- For example:
 - **Bray-Curtis dissimilarity** (presence/absence + abundance-weighted)
 - **Jaccard distance** (presence/absence)
 - **UniFrac** (phylogenetic distances: weighted/unweighted)
- These distances between samples are calculated based on the abundance and projected on the two first dimensions of a PCoA.
- other ordination methods are also used: NMDS, tSNE or simply PCA,...)



<https://pubmed.ncbi.nlm.nih.gov/30349083/>

The background features a gradient from light blue on the left to light green on the right. Overlaid on this are several concentric, wavy, semi-transparent shapes in shades of blue and green, creating a layered, organic effect.

Challenges and limitations

Challenges and limitations of metagenomics studies

- Shotgun metagenomics have higher cost than 16S amplicon, but you get higher taxonomical resolution and functional information
- Sampling biases can occur when collecting environmental samples which can also lead to misrepresentation of the microbial community
- It is also challenging to extract microbes DNA from low biomass samples - highly dependant on the sampling method and its accuracy
- Microbial genomes available and deposited in reference databases are biased towards model organisms, pathogens, species with industrial potential and cultivable bacteria
- Metagenomic sequencing can detect the presence of microorganisms but cannot easily determine if they are pathogenic (Do they carry their toxins? Are they active?)

Challenges and limitations of metagenomics studies

- Functional profiling is hindered by the lack of annotation for many genes
- “Microbial dark matter” – microbes not yet characterized can be recovered by assemble-based methods, but still many reads are not used in the assembly – sequencing noise, DNA contamination, and bacteria and plasmids that remain obscure even after some parts of their genomes are assembled
- “Live or dead” dilemma – DNA persists in the environment after the death of the host cell, sequencing results may not be representative of the active microbial population
- Snapshot of the microbiome - many metagenomics studies are cross-sectional and can not assess intra- vs inter-subject variation – more longitudinal studies are needed
- Disentangling the cause from the effect – follow ups, longitudinal and prospective studies needed
- Metagenomics have not yet reached the level of standardization of other high-throughput techniques